ME 14 Final Design Debrief

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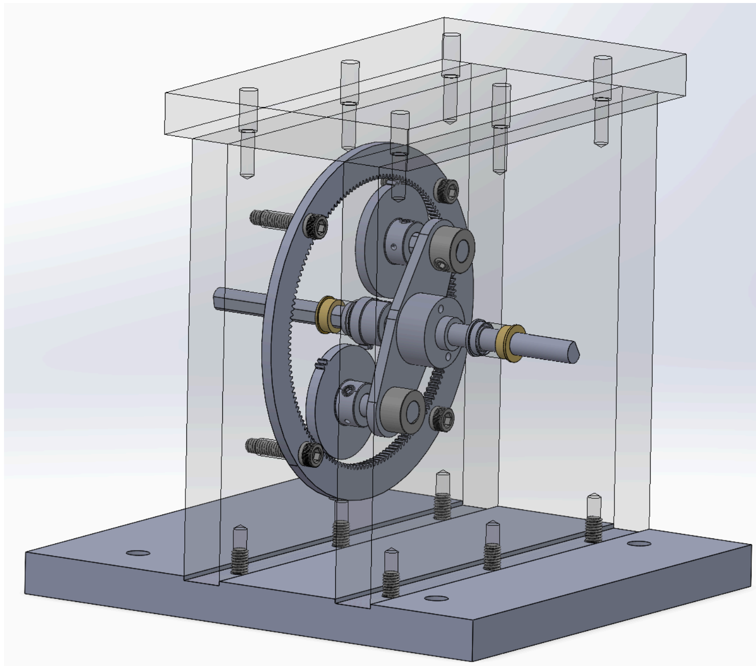

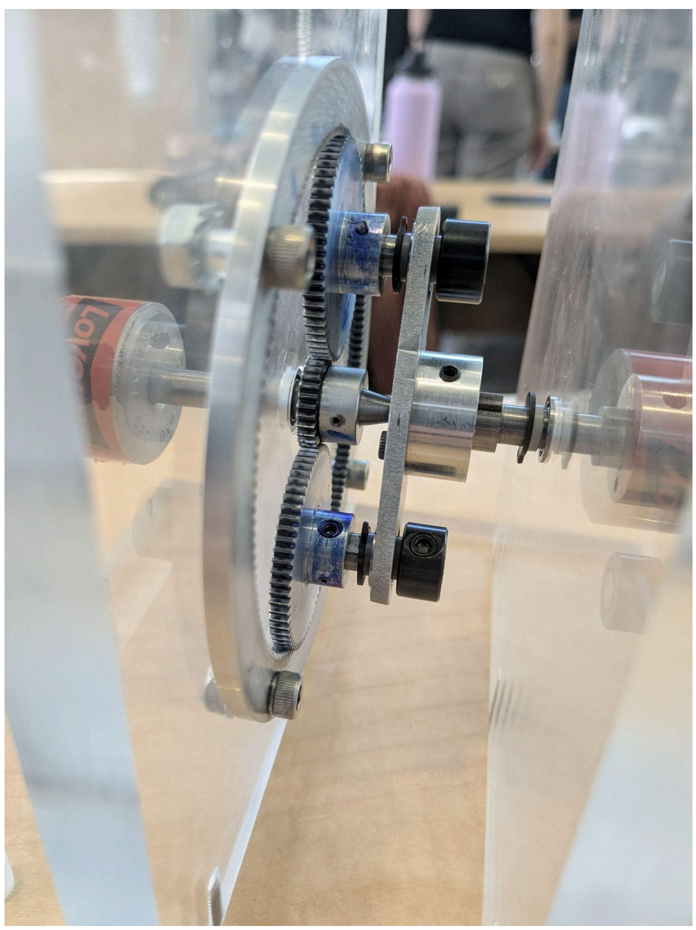

(1) Did aspects of your design perform better than you had anticipated?

The MATLAB analysis assumed 70% power transmission efficiency and a transmission ratio of 6.5. Under these parameters, we predicted a maximum angular velocity of ~466 RPM, while our maximum RPM during our first trial run was 485 as seen in Fig. 1. Additionally, the MATLAB simulation predicted a maximum power output of ~12 W, which we also surpassed with a maximum power of ~13 W; see Fig. 2.

Comparing our transmission to the planetary transmission held in the shop, we were also pleasantly surprised that our gears turned with much less resistance. This could be attributed to a variety of differences between the two systems like potential tolerancing errors, the difference in the number of planetary gears (2 versus 3), the overall mass, moment of inertia, quality of bearings, etc. Notably, the gears used in our transmission were lighter and thinner, corresponding to less overall inertia in our system.

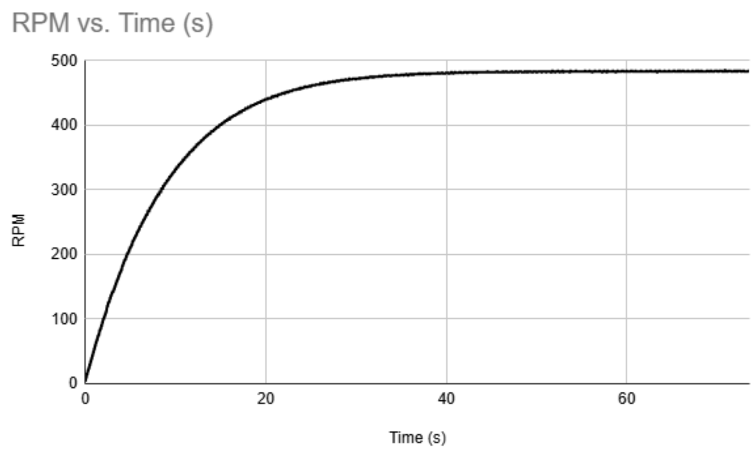

Figure 1: *RPM vs. Time (s) for our first trial run. Our max RPM was 485.*

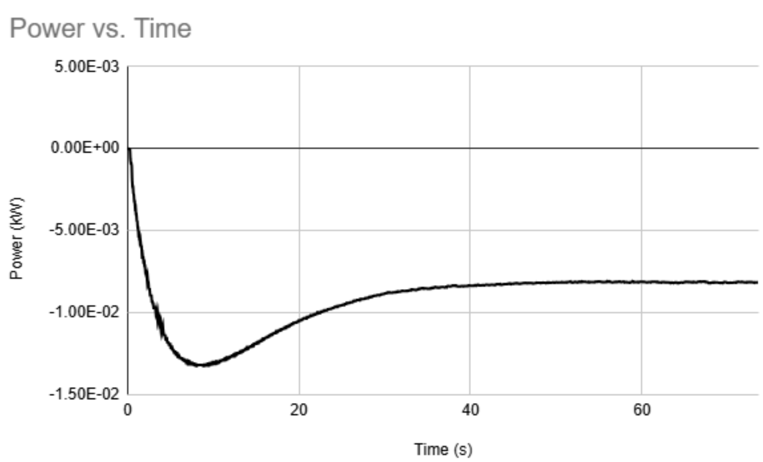

Figure 2: *Power (Kilowatts) vs. Time (s) for our first trial run. Our max power output was 0.013 Kilowatts.*

(2) Did things go wrong that you didn't expect?

We initially designed each side wall to include a flanged bearing as well as a bushing to support the input and output shafts radially and reduce deflections. It turned out, however, that the addition of the bushings introduced significant frictional losses. During testing in the rig, acceleration was reduced, and the bike wheel very quickly decelerated after the motor was turned off. Moreover, there was visible wear on the input shaft due to contact with the bushing. Upon further inspection of the design, we realized that the addition of the bushing, specifically on the input side wall, overconstrained the input shaft, as it already had two points of support from the flanged bearing mounted on the side wall and the bearing in the carrier. We ultimately decided to remove the bushings from our final assembly, which greatly improved performance and significant losses were no longer apparent.

Axial play in the test rig output flex coupler (it is free to shift about half an inch) meant that the coupler of our transmission did not always completely interface with it. This might have introduced unexpected differences in our transmission's performance as well.

Also, although not something that went *wrong* per se, the ring gear also presented an unexpected challenge as we didn't think about the logistics of accurately drilling mounting holes in the gear such that they aligned well with those on the side plate. We solved this by clamping the gear to an aluminum plate with socket head screws and using the mill to properly edge find and drill the holes.

(3) What would you have done differently were you given more time?

Given more time, we would have run more tests on our design, as we were behind schedule due to long queues for machining appliances like the mills. Additionally, we would have considered lengthening the shafts to ensure that the couplers at the ends of our transmission fully meshed with the test rig flex couplers. Ensuring complete engagement of the couplers would have helped maximize power transmission efficiency and reduced wear from misalignment or intermittent contact. Given more time, we would have refined the shaft support strategy, potentially repositioning the bearings to reduce shaft deflection, as a compromise to using additional bushing support, which ended up over-constraining the shafts. With more time, we could've also iterated on machined parts more to achieve tighter shaft and hole tolerances and performed a more meticulous assembly of the whole system to reduce the chance of uneven load distributions, vibrations, and unnecessary play. Lastly, we would have explored the effects of different lubrication amounts (spindle grease) on our system's performance.

(4) What was the score you expected to receive based on your analysis, and what score did your team receive?

Our analysis using MATLAB to optimize the transmission ratio predicted ~14.5 seconds to reach 250 RPM and a maximum angular velocity of ~466 RPM, resulting in an overall score of ~32.2. This optimization assumed a transmission power efficiency of 70% to account for losses in the bearings (5% per bearing), between gears, potential tolerance stack up, and axial and radial misalignment within the system. During the competition, our transmission achieved 250 RPM in 6.5 seconds and a maximum RPM of 485, which resulted in a final score of 76.14. Thus, our transmission performed better than expected in both metrics that make up the score, and we received a higher score than our analysis predicted as a result.

(5) Discuss what you view as the critical contributions to why you may not have received your predicted score.

Our score greatly exceeded the predicted score of the model which could be due to a conservative choice of model parameters in our analysis or the model itself being more conservative than reality. Our performance could also be attributed to our design choices to reduce misalignments. One design feature that likely contributed to this was the radial support portion of the input shaft which constrains the carrier (and output shaft) to rotate coaxially with the input shaft. This feature greatly reduced misalignments due to shaft wobble. Additionally, we assembled the acrylic housing on a granite block relative to a common datum, which helped reduce positional misalignments between different features, like the input and output bore holes on the side plates. These choices along with generally good tolerancing choices and machining produced smooth torque, power, and RPM curves with little jitter/oscillation during both trials which seems to suggest a higher-than-expected efficiency or, at the very least, that our system had few imperfections that could have caused noise in the data.